Case Study: YouTube Shorts Performance Prediction

Problem statement: The goal of this project is to leverage Supervised Machine Learning and Advanced Analytics to predict the viral potential of YouTube Shorts. The analysis aims to understand which video characteristics (duration, title, category) and publishing behaviors (upload hour) most significantly drive high engagement. The ultimate objective is to provide data-driven strategies to maximize content reach and channel monetization.
Dataset link:
https://drive.google.com/drive/folders/1Qt1HfSoTyCKiyDy2frR-hYOT9UvfwGq7?usp=sharing
Dataset description: Dataset Description.docx
Approach document: Refer to the document below to help guide you in solving the questions.

- https://docs.google.com/document/d/1BPwWz_aKu72X-1aIbMwJJ7cpFSJCuVIlirS-UZ2q9Ag/edit?tab=t.0

Business questions :

1. Calculate the Engagement Rate for all Shorts and categorize performance into Low, Medium, and High tertiles. What is the distribution of the target variable, and does it suggest any class imbalance challenges?
2. Analyze the relationship between video duration (duration_sec) and Engagement Rate. Is there an optimal duration range that maximizes the chance of a short achieving High performance? What is the model's reliance on this feature?
3. Analyze the influence of the upload_hour on average Engagement Rate. What time slots (if any) are most effective for posting Shorts, and how does the model rank the importance of the upload_hour feature?
4. Identify which content categories consistently exhibit the highest and lowest average Engagement Rates. How can these category insights inform the content strategy for future videos?
5. Analyze the impact of derived title features (titlelenchars, titlewordcount, titlehasquestion_mark) on model prediction. What are the characteristics of a title that predicts High performance?
6. Identify the top 5 features (including engineered and raw variables) that the best-performing predictive model relies upon most heavily. Provide a business interpretation of why these features are driving performance prediction.

7. Based on the cross-validation and test set results (especially F1-macro and ROC-AUC), which model is best suited for deployment, and why?
8. Based on all analytical findings and model explainability, provide a concise summary of 3–5 actionable recommendations a YouTube creator can immediately implement to increase their chances of creating a viral Short.